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Nickel sensitivity across life stages and laboratory populations in the great pond snail, *Lymnaea stagnalis*

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ABSTRACT

Nickel (Ni) is a prevalent metal in aquatic ecosystems, introduced through both industrial activities and natural processes. While Ni is an essential trace element for some organisms, elevated environmental concentrations can have toxic effects on sensitive aquatic species. This study evaluates life stage-specific Ni sensitivity in the great pond snail, *Lymnaea stagnalis*, using chronic tests of growth, reproduction, and survival across developmental stages: freshly hatched juveniles, two-week-old juveniles, adults, and laboratory populations (composed of individuals at different life stages). *L. stagnalis* is an important species in the environmental risk assessment of Ni, exhibiting high but variable sensitivity to Ni exposure.

Freshly hatched snails were the most sensitive life stage, with significant growth inhibition observed at low Ni concentrations (6.0 μg Ni L^{-1}). Two-week-old juveniles exhibited intermediate sensitivity, while adult snails exhibited no observable effects on growth or reproduction at the highest tested Ni concentration ($\geq$ 147.7 μg Ni L^{-1}). At the population level, significant reductions in density and reproduction occurred only at the highest tested Ni concentration (64.9 μg Ni L^{-1}) and only after prolonged exposure. These findings suggest that standard adult-based toxicity tests may underestimate the ecological risks of Ni to *L. stagnalis*, particularly for juvenile life stages and population-level endpoints.

This study highlights the importance of considering life stage-specific and population-level effects in environmental risk assessments for Ni and supports their consideration in chemical risk assessment more broadly.

1. Introduction

Nickel (Ni), a prevalent metal in aquatic environments, enters surface waters through a wide variety of pathways, including domestic wastewater effluents, industrial discharge, mining activities, and natural rock weathering processes (Cempel and Nikel, 2006). Although Ni has been established as an essential trace element for some organisms, such as plants and bacteria, elevated environmental concentrations may lead to adverse effects in sensitive species (Muyssen et al., 2004; Niyogi et al., 2014). The great pond snail, *Lymnaea stagnalis*, is one of the most sensitive species to chronic Ni exposure, making it a key species in environmental risk assessment (ERA) for Ni (Crémazy et al., 2020; Niyogi et al., 2014). Because *L. stagnalis* is particularly sensitive to Ni, its inclusion in toxicity datasets is consequential for the derivation of regulatory thresholds, such as those derived from species sensitivity distributions (Peters et al., 2023; Van Regenmortel et al., 2017). Currently, only one OECD guideline exists for the testing of chemicals with *L. stagnalis*, OECD Test No 243, an adult reproduction test (OECD, 2016).

While *L. stagnalis* consistently demonstrates high sensitivity to Ni, reported effect concentrations vary markedly across studies (Crémazy et al., 2020). Wild *L. stagnalis* undergoe several developmental stages before reaching hermaphroditic reproductive maturity about 60 days after embryonic cleavage. Juveniles hatch about 11 days after a clutch is laid, then pass through multiphase postlarval development, reaching male sexual maturity approximately 30 days post-hatching and female sexual maturity around 50 days post-hatching (Fodor et al., 2020). Reported EC20 values for Ni's effect on the growth *of L. stagnalis* range from < 1.3 to 98 μg Ni L^{-1} (Crémazy et al., 2018; Niyogi et al., 2014). Individual sensitivity to Ni, like other stressors, arises from interacting genetic, physiological, and environmental factors that influence growth, reproduction, and survival (Calow and Forbes, 1998; Nys et al., 2016; Peters et al., 2019). Several studies have linked variability in *L. stagnalis* sensitivity to water composition, intrinsic sensitivity of laboratory

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cultures, temperature, test duration, choice of endpoint, and the age and size of snails at test initiation (Crémazy et al., 2020; Mattsson and Crémazy, 2023). However, the large gaps between studies reporting particularly high or low sensitivity have not been fully resolved, underscoring the need to better characterize how biotic and abiotic factors shape Ni sensitivity in *L. stagnalis*. This understanding is essential for developing testing guidelines that more reliably predict Ni toxicity, particularly for sensitive species like *L. stagnalis* (Rainbow, 2002).

Furthermore, individual sensitivity can differ from population sensitivity, which integrates both the direct and indirect effects of individual-level responses as they contribute to population-level endpoints, such as population abundance and biomass (Beaudouin and Péry, 2013; Nisbet et al., 2000; Viaene et al., 2024). Accordingly, integrating individual and population-level data supports a more complete assessment of environmental risks. Hommen et al. (2015) reported that *Lymnaea* snails were the most sensitive among a variety of tested taxa when exposed to Ni in a community context, indicating that additional consideration of this sensitive species is warranted. The often non-linear relationship between individual-level and population-level sensitivity to metals has been examined across a range of freshwater organisms, reinforcing the importance of multi-level approaches to ERA (Hansul et al., 2021; Pereira et al., 2019).

This study investigates the effects of Ni exposure on *L. stagnalis* growth across life stages, comparing observed sensitivity for freshly hatched juveniles, two-week-old juveniles, adults, and laboratory populations containing a range of developmental stages. These results can provide insights into the variability of reported *L. stagnalis* sensitivity to Ni, as well as the significance of these endpoints in ERA.

2. Materials and methods

2.1. Experimental animals

Adult snails were initially obtained in 2022 from a culture at Vrije Universiteit Amsterdam, Netherlands. Animals were then maintained in an in-house culture at Ghent University, Belgium. Juvenile tests were conducted with snails hatched from this in-house culture. Before juvenile tests, egg clutches were transferred from culture tanks into petri dishes in order to observe hatching timing and select appropriately aged snails for test initiation. Adult and population toxicity tests were initiated with adult animals that had been reared at Vrije Universiteit Amsterdam and then maintained in the Ghent University culture until testing. The culture was kept in carbon-filtered Ghent University tap water, which was also used as the experimental medium for all experiments (Table 1). The culture was fed a mixture of organic butterhead lettuce and TetraMin fish flakes.

2.2. Toxicity tests

Toxicity tests to investigate the sensitivity of single life stages were conducted for adults (Section 2.2.1), freshly hatched juveniles (Section 2.2.2), and two-week-old juveniles (Section 2.2.3). All single life stage tests were 28-day chronic tests. To evaluate diet effects, different diet treatments were tested in two-week-old juveniles. Lastly, a laboratory population, composed of a range of life stages, was used to assess the impact of Ni on population density and reproduction (Section 2.2.4). To allow for the emergence of recruitment and competition dynamics, the population-level Ni exposure was extended to 133 days. In all tests, snails were exposed to a range of Ni concentrations (Table 2), and their growth, reproduction, and survival were monitored over time. Stock Ni solutions ($NiCl_2$) were prepared in acidified Milli-Q water and diluted with Ghent University tap water to the test-specific concentrations used as the experimental medium. Ni concentrations were selected to span the range of sensitivities reported in the literature and summarized by Crémazy et al. (2020) (Table 4).

Table 1
Water quality parameters (mean ± standard deviation) as observed for each test.

	Freshly Hatched	Two-Week Fish Flakes	Two-Week Lettuce	Adult	Population
pH	8.27 ±0.19	8.04 ±0.23	8.03 ±0.16	8.03 ±0.58	8.03 ±0.31
DO [mg L^{-1}]	8.60 ±0.20	8.32 ±0.16	8.27 ±0.25	7.64 ±0.59	7.62 ±0.80
DOC [mg C L^{-1}]	1.73 ±0.24	1.54 ±0.28	1.42 ±0.22	5.59 ±0.93	3.45 ±0.78
Na [mg L^{-1}]	46.74 ±0.78	56.99 ±1.95	55.66 ±1.85	26.87 ±2.45	45.36 ±2.89
Ca [mg L^{-1}]	52.04 ±5.44	59.05 ±6.66	58.63 ±6.64	51.53 ±15.16	55.92 ±1.67
K [mg L^{-1}]	5.35 ±0.22	6.4 ±0.61	9.07 ±1.83	13.89 ±10.81	5.24 ±0.22
Mg [mg L^{-1}]	8.38 ±0.14	10.18 ±0.73	10.28 ±0.71	6.88 ±1.09	8.55 ±0.32
Chloride[a] [mg L^{-1} as Cl^-]	59.01 ±2.20	59.01 ±2.20	59.01 ±2.20	39.98 ±1.73	49.07 ±1.97
Sulfate[a] [mg L^{-1} as SO_4^{2-}]	70.27 ±2.29	70.27 ±2.29	70.27 ±2.29	51.21 ±13.75	72.24 ±5.25
Ammonium[b] [mg L^{-1} as NH_4–N]	0.20 ±0.15	-	-	1.41 ±0.54	0.41 ±0.31

[a] measured only in fresh medium.
[b] measured only in old medium.

2.2.1. Toxicity test for adult snails

Adult exposure was designed to follow OECD Test No. 243 as closely as possible (OECD, 2016). Five experimental Ni concentrations were tested, in addition to a control treatment: 6.2, 12.7, 30.6, 59.8, 147.7 μg Ni L^{-1} (mean dissolved concentrations; weekly measurements are provided in SI Table 1). Adult exposure was conducted in 1 L Weck jars, with six replicates per treatment, each jar initially containing five adult snails with a minimum shell length of 2.5 cm. Per guidelines for the *L. stagnalis* reproduction test, adult snails were housed in groups of five to allow for sexual reproduction. Snails were allowed an acclimation period of two weeks in the test vessels, during which control test validity criteria (reproduction and survivorship) were confirmed. Following the acclimation period, Ni exposure was conducted for 28 days.

Adult snails were fed organic butterhead lettuce five times per week. Lettuce was cut into circular pieces, with a uniform surface area, to allow for later estimation of ingestion. At each feeding, 141 cm^2 of lettuce was placed in each jar of five snails, equivalent to 20 cm^2 of lettuce per snail per day, which is approximately ad libitum feeding for the snails (portions were not fully consumed). The medium in each jar was entirely replaced with fresh medium three times per week, with samples taken of spent and fresh medium for physicochemistry analysis (Section 2.3). At this time, any remaining lettuce from previous feedings was also removed.

With each medium replacement (three times per week), any mortality was recorded and dead snails were removed. The number of egg clutches in each jar was counted, and all egg clutches were stored for later analysis. Subsequently, each clutch was scanned, and the total number of eggs in each clutch was counted. Snail shell length was measured once per week using digital calipers. The growth of individual snails within each jar was monitored via the placement of numbered markers, originally intended for bees, on the shell of each animal (BeeFun, Apis International, Calgary, Canada). After length measurement, each snail was blotted dry, and the whole-animal wet weight was recorded. Once per week, lettuce remaining from the previous feeding was removed, scanned, and the remaining surface area was estimated using the "Measure" feature in ImageJ (available at http://rsb.info.nih.gov/ij).

Table 2
Dissolved Ni concentration (mean ± standard deviation) for each treatment in all toxicity tests. Dissolved Ni by week and nominal Ni concentrations for each treatment can be found in Supplemental Information.

	Control [μg Ni L^{-1}]	Treatment 1 [μg Ni L^{-1}]	Treatment 2 [μg Ni L^{-1}]	Treatment 3 [μg Ni L^{-1}]	Treatment 4 [μg Ni L^{-1}]	Treatment 5 [μg Ni L^{-1}]	Treatment 6 [μg Ni L^{-1}]
Freshly Hatched	1.9 ± 1.0	3.4 ± 0.4	6.0 ± 0.4	12.7 ± 1.0	30.6 ± 1.5	82.0 ± 17.2	198.6 ± 19.3
Two-Week-Old Fish Flakes	1.1 ± 0.3	3.1 ± 0.6	5.7 ± 0.6	13.0 ± 0.6	31.9 ± 2.54	77.4 ± 6.9	201.2 ± 21.2
Two-Week-Old Lettuce	1.1 ± 0.3	2.9 ± 0.6	5.3 ± 0.9	12.2 ± 0.7	29.9 ± 1.9	73.7 ± 5.1	194.1 ± 15.6
Adult	0.8 ± 0.4	6.2 ± 1.0	12.7 ± 3.3	30.6 ± 5.5	59.8 ± 9.1	147.7 ± 19.1	-
Population	1.0 ± 0.6	2.2 ± 1.0	7.2 ± 1.5	64.9 ± 12.0	-	-	-

2.2.2. Toxicity test for freshly hatched snails

Before exposure began, clutches that appeared near-to-hatching were transferred from culture tanks into petri dishes for hatching observation. Upon hatching, snails that hatched within a 24-hour period of each other were transferred into individual 0.5 L Weck jars filled with exposure medium. Six experimental Ni concentrations were tested, in addition to a control treatment: 3.4, 6.0, 12.7, 30.6, 82.0, 198.6 μg Ni L^{-1} (mean dissolved concentrations; weekly measurements are provided in SI Table 1). A wider range of Ni concentrations was chosen for juvenile testing based on a lack of observed effects in adult testing. Once snails were moved into the exposure medium, Ni exposure was conducted for 28 days.

Snails were fed three times per week, with each snail receiving the equivalent of 0.43 mg of TetraMin fish flakes per individual per day, which is greater than documented ad libitum feeding in existing studies (Berteloot et al., 2015; Zimmer et al., 2012). A lettuce diet was not investigated for the freshly hatched age group, as previous studies have indicated that fish flakes are a better food source for juvenile snails than lettuce, possibly owing to their relatively high protein content (Zimmer et al., 2012). Once per week, prior to medium replacement and feeding, spent medium was sampled for physicochemistry (Section 2.3). Spent medium was then replaced entirely with fresh medium. Fresh medium was sampled at batch preparation, with ≥ 24 h between Ni spiking and sampling.

Survival was monitored three times per week, with feeding. Once per week, snail size (shell length) was measured using a stereomicroscope (Olympus SZX10) and a camera (Olympus UC30).

2.2.3. Toxicity test for two-week-old snails

The same procedure described for freshly hatched snails was followed for two-week-old snails, except for the dietary regimen. For two-week-old snails, two different diet treatments were used. One group of snails was fed fish flakes, as described for freshly hatched snails, while the other was fed lettuce. Lettuce-fed snails were provided with a 19.6 cm^2 portion of lettuce at each feeding, which is approximately the ad libitum requirement reported for adult *L. stagnalis*. Six experimental Ni concentrations were tested, in addition to a control treatment: 3.0, 5.5, 12.6, 30.9, 75.6, 197.7 μg Ni L^{-1} (mean dissolved concentrations across diet treatments; weekly measurements per diet treatment are provided in SI Table 1).

2.2.4. Toxicity test for laboratory populations

To investigate the implications of Ni toxicity to *L. stagnalis* at the laboratory population level, three experimental Ni concentrations were tested in addition to a control treatment: 2.2, 7.2, and 64.9 μg Ni L^{-1} (mean dissolved concentrations; weekly measurements are provided in SI Table 2), with five replicates per treatment. Before exposure began, ten adult snails of reproductive age were transferred into each of the 20 L tanks for an acclimation period. Twice per week, half of the tank medium was removed and replaced with fresh medium. Once per week, before medium renewal, spent medium was sampled from each tank for physicochemistry analysis (Section 2.3). A composite sample was also taken for each of the treatments once weekly, before the second medium renewal. Snails were fed lettuce three times per week, with each tank receiving 35 g per week of lettuce. This quantity reflects ad libitum feeding for ten adult snails, and it was observed that snails did not consume the entire portion during the acclimation period. Large, noticeable pieces of excess food were removed from the tank during medium renewal. Ni exposure began once tanks produced ≥ 10 egg clutches per week (~1 clutch per snail, in alignment with OECD Test No 243). Upon beginning Ni exposure, any existing egg clutches were removed from the tanks, so only ten adult snails remained at test initiation. Unlike in adult life stage testing, no egg clutches or juvenile snails were removed from the tanks once exposure began. Population-level exposure continued for 133 days.

Survival was monitored twice per week, with medium renewal. Once per week, the number of new clutches produced in each tank was monitored. Clutches were marked on the exterior of the tank to ensure that new clutches could be identified the following week. Every two weeks, the population density of each tank was estimated. After medium renewal, when water was clearest, all four sides of each tank were photographed. These photographs were then imported into ImageJ, where snail size was measured. Scale in each photograph was established using a physical measuring grid which was fixed to the exterior of the tank. Snails with an estimated shell length ≥ 5 mm were counted as an indicator of relative population density. Based on control observations during juvenile testing, snails likely reached this length approximately 5 weeks after hatching. Smaller individuals were not counted, as they could not be consistently identified in photographs. As such, the density of small juvenile snails in the population is not reflected in these estimates.

2.3. Physicochemical analysis

Measurements of pH, dissolved oxygen (DO), and conductivity were taken from fresh and spent medium before medium replacement (Table 1). For both fresh and spent medium, pH was measured using a pH glass electrode (P407, Consort), conductivity was measured using a WTW Cond 315i meter, and DO was measured using a WTW Oxi 3210 m

Samples for total and dissolved Ni, other dissolved metals (Cr, Mn, Fe, Co, Cu, Zn, Pb) and major cations (Na, Mg, K, Ca), total organic carbon (TOC), and dissolved organic carbon (DOC) were collected from fresh medium at batch preparation (≥ 24 h after Ni spiking) and from spent medium at least once weekly before medium replacement (Table 1, Table 2). Dissolved fractions were obtained by filtration through 0.45-μm PES filters (Novolab, Belgium). Metal and cation samples were acidified to 1 % using trace-metal-grade HNO_3 (Pico Pure, Chem-Lab, Zedelgem, Belgium) immediately upon sampling and refrigerated at 1–4 °C until analysis by ICP-OES or ICP-MS. Treatments with nominal Ni concentrations > 5.0 μg Ni L^{-1} were analyzed by ICP-OES, with limits of detection and quantification for Ni of 1.2 and 5.0 μg L^{-1}, respectively (Thermo Scientific, Waltham, MA, United States). Treatments with nominal Ni concentrations ≤ 5.0 μg Ni L^{-1} were analyzed by ICP-MS (Agilent 7700x, Agilent Technologies, Santa Clara, CA, United States), with a limit of quantification of 0.5 μg Ni L^{-1}. TOC and DOC were analyzed on a Shimadzu TOC-L analyzer (Kyoto, Japan). Chloride and sulfate in fresh medium (at batch preparation) and ammonium in spent medium (at medium replacement) were measured by spectrophotometry (Aquamate, Thermo Electron Corporation; chloride: Merck Spectroquant 1.14897.001; sulfate: Merck Spectroquant

1.14548.001; ammonium: Hach LCK304).

2.4. Statistical analysis

Snail growth was defined as the percent increase in shell length from the length at test initiation (*$Length_0$*) to the length (*$Length_t$*) on a given exposure day (*t*).

$$\text{growth}(t) = ((\text{Length}_t - \text{Length}_0) / \text{Length}_0) \times 100 \quad (1)$$

NOEC and LOEC values for individual snail growth and reproduction were determined using a Kruskal-Wallis test followed by a one-sided Conover post hoc test with Bonferroni-Holm adjustment, comparing each treatment with its corresponding control ($\alpha = 0.05$). NOEC and LOEC determinations for individual survival data were made using a one-sided Fisher's exact test with Bonferroni-Holm adjustment ($\alpha = 0.05$).

Population density and reproduction were analyzed using negative binomial generalized linear mixed models with a log link. Fixed effects were Ni treatment, day, and their interaction, and tank was included as a random intercept. Because day was treated as a categorical fixed effect, all inferences are within-day. For each observation day, a one-sided Dunnett's test compared the model-estimated marginal mean for each treatment with that day's control. Day-specific NOEC and LOEC were determined from Dunnett-adjusted p-values ($\alpha = 0.05$). Reproduction for each tank was analyzed as tank-level clutch production during the observation interval and as clutches produced per surviving snail per day during the observation window. Given the repeated observations over an extended exposure period, population-level NOEC and LOEC were deemed reliable only if a significant effect was observed for at least two consecutive observations (Hommen et al., 2015).

Growth and survival effect concentrations (ECx/LCx) were estimated using the Toxicity Relationship Analysis Program (TRAP v1.30a, Duluth, MN, United States). For each endpoint, EC10/20/50 estimates for growth and LC10/20/50 estimates for survival were derived (Table 3). Prior to model fitting, exposure concentrations were log-transformed. Growth effect concentrations were estimated using a 3-parameter log-logistic model. Lethal effect concentrations were estimated using a 3-parameter triangular tolerance distribution analysis. Some exposure-response fits showed inadequate partial effects; these cases are flagged and should be interpreted as exploratory. All statistical TRAP outputs, including model parameters, are presented in Supplemental Information (SI Outputs 1–8). No dose-response analysis was performed for population endpoints.

3. Results

3.1. Individual-level sublethal effects

3.1.1. Freshly hatched snails

When exposure was initiated with freshly hatched snails ($\leq$ 24 h post-hatching), Ni-exposed snail growth showed significant decreases relative to controls at the two highest tested concentrations (82.0 and 198.6 μg Ni L^{-1}), beginning from the first post-exposure measurement (day 7; SI Fig. 1). After 14 days of exposure, significant reductions in snail growth, relative to controls, were observed in all but the lowest tested concentration (3.4 μg Ni L^{-1}), with significant effects at 6.0 μg Ni L^{-1} or more (SI Fig. 2). The NOEC of 3.4 μg Ni L^{-1} and LOEC of 6.0 μg Ni L^{-1} were maintained throughout the remainder of the exposure period (Fig. 1). However, at the conclusion of exposure (day 28), the 3.4 μg Ni L^{-1} group showed a borderline-significant decrease in growth ($p = 0.058$). Given the high background Ni concentration (1.9 μg Ni L^{-1}) in the freshly hatched test, together with the estimated effect concentrations from dose-response analysis (SI Output 2), this result can be plausibly interpreted as an unbounded LOEC of $\leq$ 3.4 μg Ni L^{-1}, depending on the significance criteria. We use 3.4 μg Ni L^{-1} as the NOEC for freshly hatched snails in comparisons across life stages to maintain methodological consistency (Table 3).

Table 3
NOEC, LOEC, and ECx estimates for L. *stagnalis* Ni sensitivity, derived from each of the toxicity tests. ECx estimates are presented with 95 % confidence intervals (corresponding dose-response curves can be found in Supplemental Information).

	Freshly Hatched	2 Week Old Fish Flakes	2 Week Old Lettuce	Adult	Population
NOEC [μg Ni L^{-1}]	3.38	3.06	5.31	≥ 147.71	7.19
LOEC [μg Ni L^{-1}]	5.97	5.72	12.24	NA	64.91
EC10 [μg Ni L^{-1}]	0.11[a] (NA)	1.57 (0.60-4.11)	3.92 (2.32-6.61)	167.00[a] (NA)	–
EC20 [μg Ni L^{-1}]	0.31[a] (NA)	2.51 (1.20-5.25)	5.10 (3.41-7.63)	142.72[a] (NA)	–
EC50 [μg Ni L^{-1}]	1.8[a] (NA)	5.60 (3.70–8.47)	8.01 (6.20-10.35)	130.19[a] (NA)	–
LC10 [μg Ni L^{-1}]	64.0 (11.7–351.0)	28.6 (3.33-249.3)	138.2[a] (NA)	–	
LC20 [μg Ni L^{-1}]	94.4 (32.5-274.7)	45.5 (10.2-201.7)	163.9[a] (NA)	–	–
LC50 [μg Ni L^{-1}]	204.4 (88.9-470.0)	113.7 (43.9-294.3)	229.9[a] (NA)	–	–

[a] Inadequate partial effects; estimated concentrations should be considered exploratory. See TRAP outputs in Supplemental Information.

3.1.2. Two-week-old snails

When exposure was initiated with approximately two-week-old snails (12–14 days post-hatching) fed a fish flake diet, significant growth reductions were observed beginning from the first post-exposure measurement (day 7) at 75.6 μg Ni L^{-1} or more (SI Fig. 5), with significant reductions observed at 12.6 μg Ni L^{-1} or more after 14 days (SI Figure 6). After 21 days of exposure, growth was significantly reduced at 5.5 μg Ni L^{-1} or more, which persisted for the remainder of the exposure period, yielding a NOEC of 3.0 μg Ni L^{-1} and a LOEC of 5.5 μg Ni L^{-1} for two-week-old snails fed a fish flake diet (Fig. 1, Table 3).

When two-week-old snails were fed a lettuce diet, significant growth reductions were observed at 30.9 μg Ni L^{-1} or more beginning from day 7 (SI Fig. 5). After 14 days of exposure, significantly decreased growth was observed at 12.6 μg Ni L^{-1} or more (SI Figure 6), which persisted until the end of the exposure period, yielding a NOEC of 5.5 μg Ni L^{-1} and a LOEC of 12.6 μg Ni L^{-1} for two-week-old snails fed a lettuce diet (Fig. 1, Table 3).

Comparing control treatments, lettuce-fed control snails achieved greater shell length than fish flake-fed controls by day 28 (SI Outputs 3–4).

3.1.3. Adult snails

When exposure was initiated with reproductive adult snails (approximately six months old), no significant effects on length, weight, reproduction (clutches produced or eggs per clutch), or food ingestion were observed at any tested concentration, resulting in 28-day NOECs of

Table 4
Conditions of various chronic Ni toxicity tests measuring growth inhibition in L. *stagnalis* along with corresponding EC20 and EC50s (dissolved Ni concentration, in μg Ni L^{-1}). Adapted from Crémazy et al. 2020.

	Present Study	Present Study	Present Study	Crémazy et al. 2018	Niyogi et al. 2014	Nys et al. 2016	Schlekat et al. 2010
Snail origin	Ghent University culture (established with VU eggs)	Ghent University culture (established with VU eggs)	VU, Amsterdam	University of British Columbia	University of Miami	VU, Amsterdam eggs	University of Miami eggs
Water T (°C)	20	20	20	25 ± 1	25 ± 1	20	24 ± 2
Food	Fish flakes, ad libitum	Lettuce or Fish Flakes, ad libitum	Lettuce, ad libitum	Romaine lettuce, ad libitum	Romaine lettuce, ad libitum	Organic butterhead lettuce, 25 mg L^{-1}	Carrot, romaine lettuce, sweet potato, ad libitum
Duration	28 d	28 d	28 d	14 d	21 d	28 d	30 d
Initial age	< 24 h	13–14 d	6 month	2- 3 wk	7–8 d	2–3 d	<24 h
Measurement	Shell length	Shell length	Wet weight	Wet weight	Wet weight	Shell length	Wet weight
EC20	NA	5.1 (3.4- 7.6)[h] 2.5 (1.2- 5.3)[i]	NA	98 (73- 120)	<1.3	34 (21–54)[a] 13 (8.7- 20)[b]	1.6 (1.4- 2.3)[c] 27 (12–31)[d] 21 (6.9–62)[e] 6.3 (1.3–14)[f] 1.9 (1.7- 3.0)[g]
EC50	NA	8.0 (6.2–10.4)[h] 5.6 (3.7–8.5)[i]	NA	160 (140- 190)	1.5	> 39[a] 42[b]	6.2 (2.7–7.2)[c] 40 (34- 44)[d] 78 (66- 85)[e] 15 (8.2- 20)[f] 23 (12–36)[g]

(a) Artificial water pH 8.1, (b) Artificial water pH 8.6, (c) Calapooia River, (d) S. Platte River, (e) S. Platte River- pH Amended,.
(b) (f) S. Santiam River, (g) Zollner Creek, (h) Lettuce diet, (i) Fish flake diet.

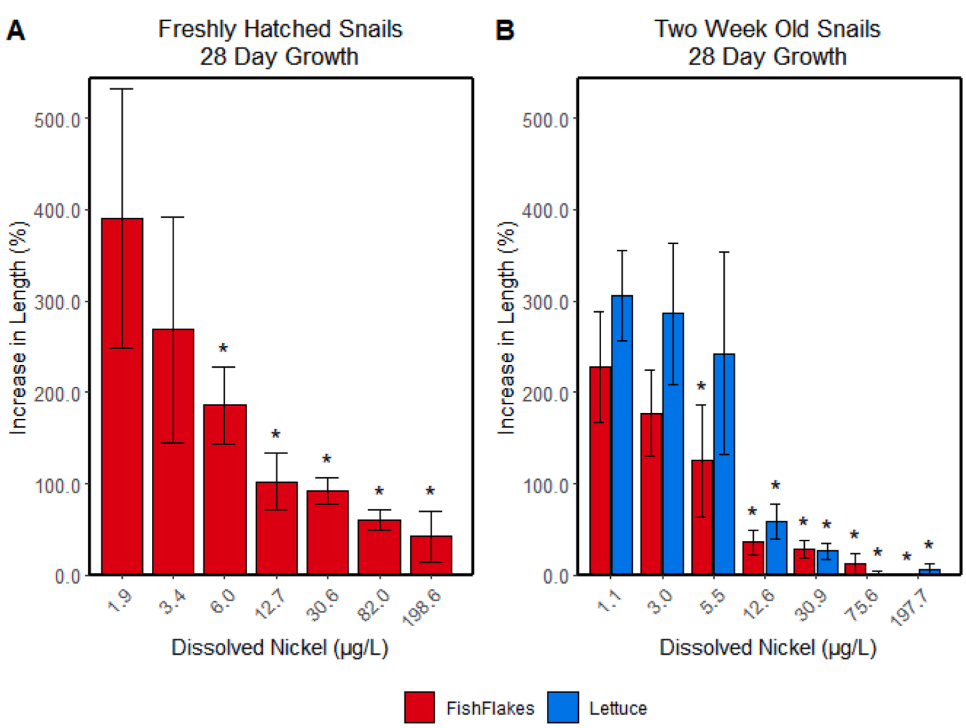

Fig. 1. Percent increase in shell length after 28 days of exposure for juvenile snails ≤ 24 h old (A) or 12–14 days old (B) at test initiation, across dissolved nickel concentrations (μg L^{-1}). Bar color indicates diet: fish flakes (red) or lettuce (blue). Bars = mean; error bars = SD. Asterisks mark concentrations with significantly lower growth than the diet-specific control (Kruskal-Wallis followed by Conover's test, one-sided; Bonferroni-Holm adjusted $p \leq 0.05$).

≥ 147.7 μg Ni L^{-1} for all endpoints (Fig. 2, Table 3).

3.2. Individual-level lethal effects

No statistically significant effects on survival were observed at any tested concentration in any life stage. The 28-day individual-level toxicity tests resulted in NOECs of ≥ 198.6 μg Ni L^{-1} for freshly hatched snails, ≥ 197.7 μg Ni L^{-1} for two-week-old snails, and ≥ 147.7 μg Ni L^{-1} for adult snails. However, for two-week-old snails fed a fish flake diet, mortality was borderline-significant in the highest tested Ni concentration of 197.7 μg Ni L^{-1} ($p = 0.051$).

3.3. Population-level effects

3.3.1. Population density

In laboratory populations, significant effects on population density (individuals > 5 mm) were observed repeatedly at the highest of the three tested concentrations (64.9 μg Ni L^{-1}) beginning from day 119 of

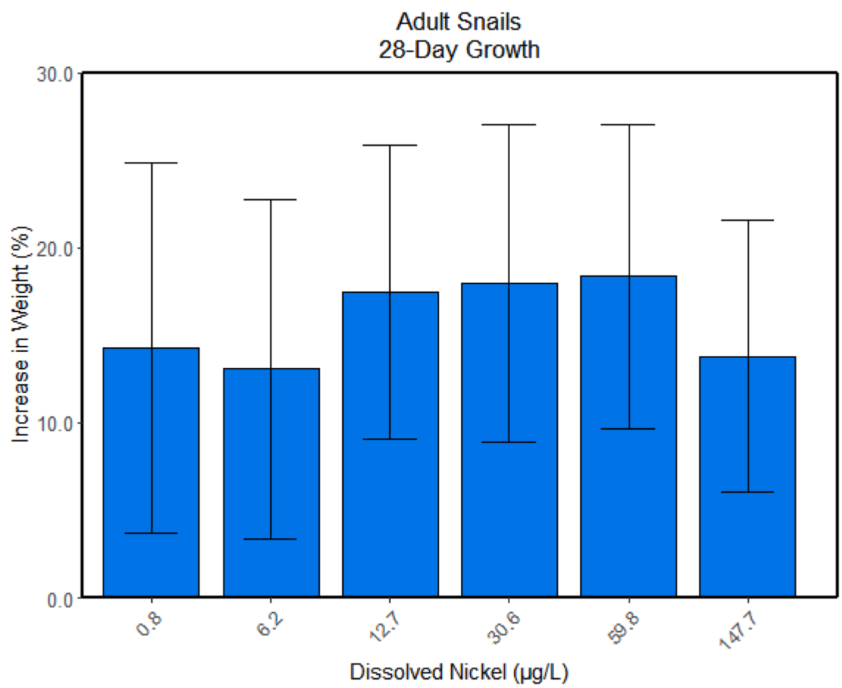

Fig. 2. Percent increase in adult whole-organism weight after 28 days of exposure for adult snails, across dissolved nickel concentrations (µg L^{-1}), fed fish flakes (red) or lettuce (blue). Bars = mean; error bars = SD. No concentrations showed growth significantly lower than the control (Kruskal-Wallis followed by Conover's test, one-sided; Bonferroni-Holm adjusted $p \leq 0.05$).

exposure and persisting through the end of testing (Fig. 3). A reduction in population density was also observed at the second-highest concentration (7.2 µg Ni L^{-1}). However, this was observed only on day 133 of exposure, at the conclusion of testing. As population-level effects required two consecutive observations to be considered reliable (Hommen et al., 2015), these results established a population density NOEC of 7.2 µg Ni L^{-1} and LOEC of 64.9 µg Ni L^{-1} (Table 3).

3.3.2. Population reproduction

Population reproduction during each observation window was

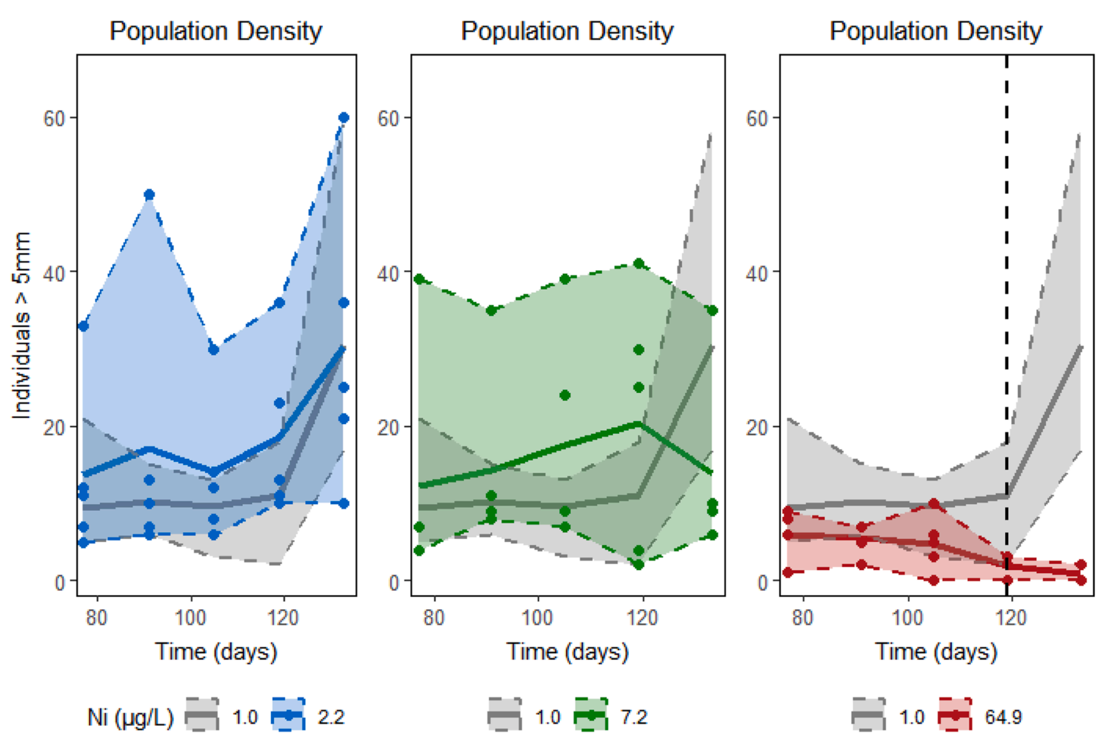

Fig. 3. Observed number of snails > 5 mm (shell length) per tank over time. Each Ni treatment is shown against the control (grey) for reference. Points show tank counts; solid lines show the treatment mean; shaded ribbons span the observed range (min-max). The vertical dashed line marks the first time point at which the treatment population density was significantly lower than the control (Dunnett test, one-sided; single-step adjusted $p \leq 0.05$). Significant effects were only detected at the highest concentration.

analyzed both as tank-level clutch production and as clutch production per surviving snail within each tank. Using either endpoint, statistically significant reduction in population reproduction was observed only at the highest tested concentration. At 64.9 μg Ni L^{-1}, tank-level clutch production was significantly decreased beginning from day 41 of exposure (Fig. 4), and clutch production per surviving snail was significantly decreased from day 48 (Fig. 5). These results established a population reproduction NOEC of 7.2 μg Ni L^{-1} and LOEC of 64.9 μg Ni L^{-1} (Table 3).

4. Discussion

4.1. Differences in life stage sensitivity

In single life stage tests, adult *L. stagnalis* snails exhibited lower sensitivity to Ni than juvenile snails. All tests used snails from the same culture, in the same medium, and with comparable diets where applicable. Freshly hatched snails were the most sensitive, followed by two-week-old snails. These results align with existing literature on Ni toxicity in *L. stagnalis*, as studies reporting the lowest Ni effect concentrations for *L. stagnalis* are those that began exposure with younger snails (Crémazy et al., 2020).

Apart from possible intrinsic life stage-sensitivity differences, multiple processes may contribute to the recurring observation that younger snails demonstrate Ni-toxicity effects at lower concentrations, as observed in the present study. Younger snails are typically smaller and have higher surface-to-volume ratios, which may influence Ni uptake and elimination dynamics (Crémazy et al., 2020; Mattsson and Crémazy, 2023). At test initiation, freshly hatched snails had an average shell length of 1.4 mm, compared to 2.4 mm for two-week-old juveniles and 28.3 mm for adults. Although the surface-to-volume ratio of soft tissues is likely most relevant to uptake and elimination dynamics, growth in shell length correlates closely with total body weight in *L. stagnalis* (Arambašic et al., 2013; Pinkina et al., 2022). Furthermore, the relationship between shell length and soft tissue weight is consistent with approximately isomorphic growth in post-embryonic *L. stagnalis* (Zonneveld and Kooijman, 1989). Smaller snails, with higher surface-to-volume ratios, may take up Ni more rapidly. This mechanism may contribute to the relatively high sensitivity observed in younger snails, and could be compounded if Ni regulatory or excretory functions are intrinsically limited in earlier developmental stages (Crémazy et al., 2020).

Several experimental factors complicate a clear delineation of possible intrinsic sensitivity differences between the life stages examined in this study. Juvenile snails were housed individually from the outset to enable tracking of individual growth over time. In contrast, adult snails were housed in groups of five to allow for sexual reproduction. The population-level experiment also necessarily allowed for interactions among individuals. Differences in population density and social isolation dynamics are known to influence growth and morphology in a variety of taxa (Burns, 2000; Ojelade et al., 2022). In *L. stagnalis,* conspecific interactions, through pheromones, excretions, or mating, have been shown to influence both growth and reproductive output (Koene and Ter Maat, 2004; Nakadera et al., 2015). Social rearing context also influences allocation of resources to male versus female reproductive roles (Koene and Ter Maat, 2004). Differences in rearing conditions across exposure tests may have prompted differences in snail growth and/or reproduction, independent of Ni exposure. Any such differences could also affect the observed Ni toxicity effects, complicating comparisons across life stages.

The bioavailability of Ni may also have differed between life stage tests, contributing to observed sensitivity differences. The same test medium was used across all life stage experiments; however, higher animal density in adult test vessels, together with associated feeding, resulted in higher DOC than in juvenile tests (Table 1). The average DOC in adult testing was 5.6 mg L^{-1}, whereas the average DOC concentrations for freshly hatched and two-week-old juveniles were 1.7 mg L^{-1} and 1.5 mg L^{-1}, respectively. Previous studies have shown that elevated

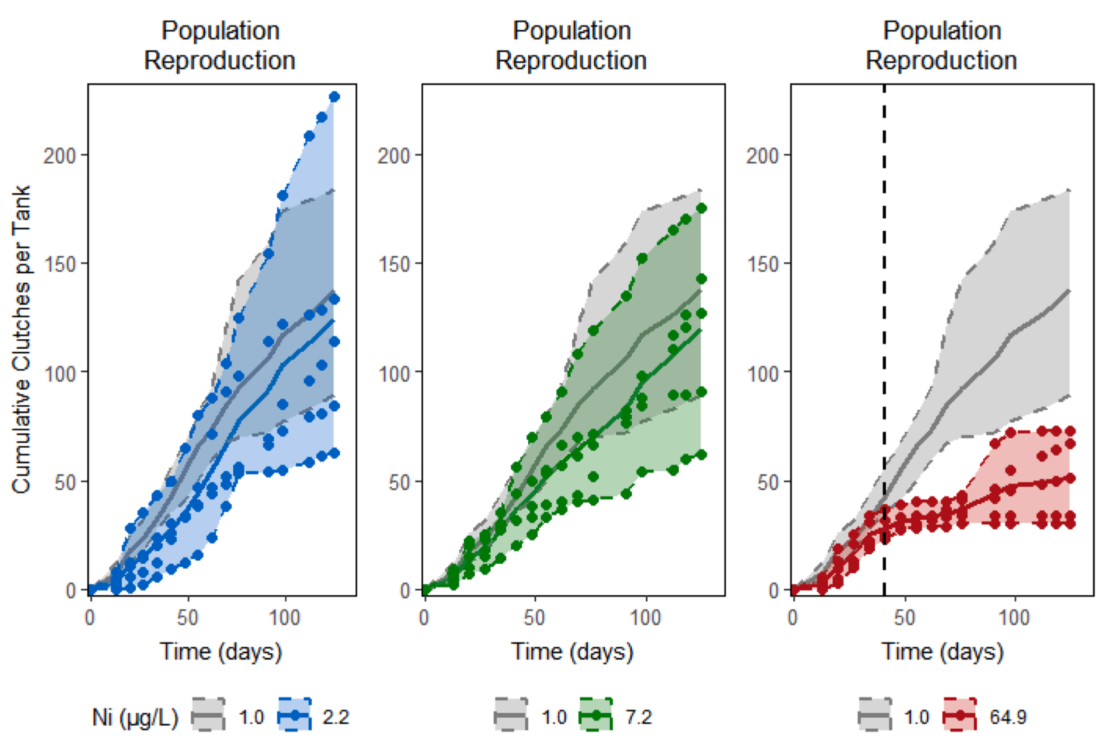

Fig. 4. Observed number of egg clutches (plotted cumulatively) per tank over time. Each Ni treatment is shown against the control (grey) for reference. Points show tank counts; solid lines show the treatment mean; shaded ribbons span the observed range (min-max). The vertical dashed line marks the first time point at which the treatment tank-level clutch production (within the observation interval) was significantly lower than the control (Dunnett test, one-sided; single-step adjusted $p \leq 0.05$). Significant effects were only detected at the highest concentration.

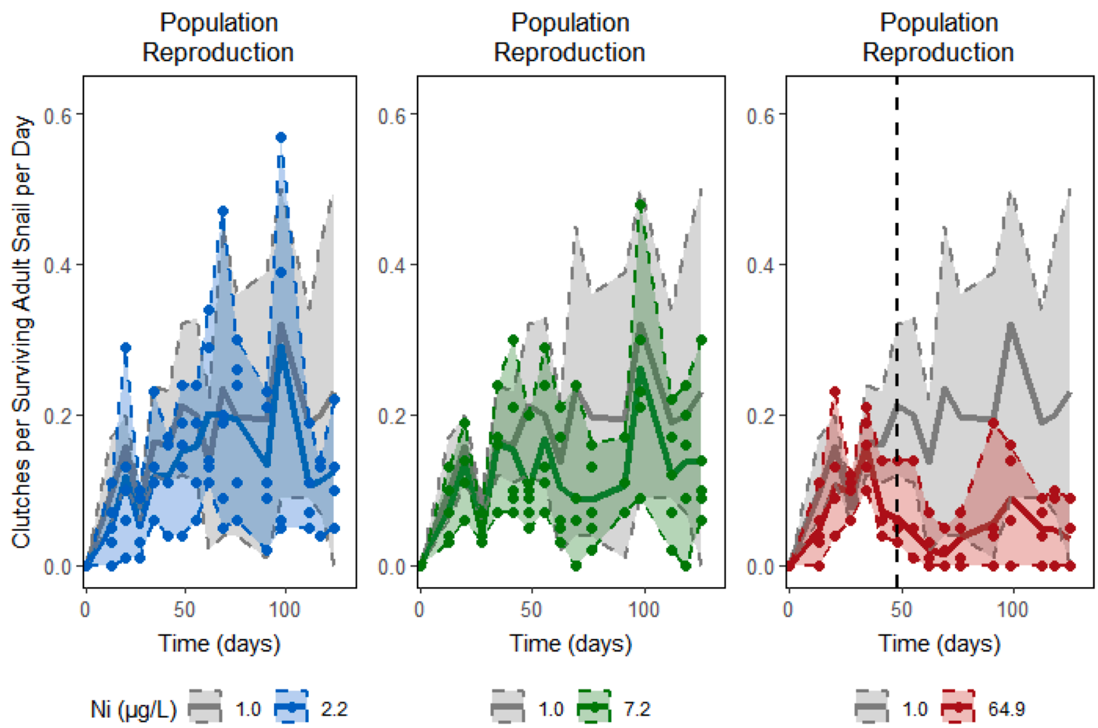

Fig. 5. Observed number of egg clutches (per surviving adult snail, per day) over time. Each nickel treatment is shown against the control (grey) for reference. Points show tank counts; solid lines show the treatment mean; shaded ribbons span the observed range (min-max). The vertical dashed line marks the first time point at which the treatment clutch production (per surviving adult snail, per day) was significantly lower than the control (Dunnett test, one-sided; single-step adjusted $p \leq 0.05$). Significant effects were only detected at the highest concentration.

DOC can mitigate Ni toxicity in *L. stagnalis*, leading to reduced Ni bioaccumulation and diminished effects on growth and survival (Custer et al., 2016). Although Custer et al. (2016) examined higher waterborne Ni concentrations (~400 μg Ni L^{-1}), attenuation was observed when DOC increased from ~2–4 mg L^{-1} (concentrations comparable to this study) to ~10 mg L^{-1}. Biotic ligand models have also been used to reduce uncertainty surrounding Ni effect concentrations for *L. stagnalis* in a range of natural waters with Ni and DOC concentrations similar to those examined here (Schlekat et al., 2010). Taken together, these findings are consistent with the possibility of greater DOC-mediated Ni attenuation in adult tests. Although elevated DOC in adult exposures may have reduced Ni bioavailability, no effects were observed in adults at any test concentration, even though the highest tested concentration was approximately 43-fold higher than the freshly hatched NOEC (Table 3). Therefore, the < 4-fold difference in DOC between tests is unlikely to fully account for the observed difference in sensitivity.

Ammonium (measured as NH_4—N) concentrations also varied between life stage tests (Table 1), likely due to animal density and food inputs. The adult test had higher ammonium concentrations than juvenile (freshly hatched) or population tests. Total ammonia as nitrogen (TAN) was estimated from measured NH_4—N using concurrent pH measurements (U.S. EPA, 2013). In the adult test, weekly TAN estimates occasionally exceeded the EPA 2013 chronic (30-day) freshwater criterion at pH 8 and 20 °C (~1.7 mg TAN L^{-1}). However, over the 28-day exposure, the average concentration (1.5 mg TAN L^{-1}) did not exceed the chronic criterion. Total ammonia was never estimated to exceed the EPA four-day average criterion (2.5 × the chronic criterion). Estimated total ammonia also remained well below the species-specific chronic value reported for *L. stagnalis* (> 8 mg TAN L^{-1} at pH 8.25) (Besser et al., 2016). No lethal or sublethal effects were observed at any Ni concentration for adult snails, consistent with negligible ammonia-related stress. For all other tests, total ammonia remained within EPA Ambient Water Quality Criteria (AWQC) chronic guidelines, and there were no significant Ni treatment effects on ammonia concentrations.

4.2. Effects of diet on snail growth

L. stagnalis fed a fish flake diet were more sensitive to Ni toxicity (lower NOEC) than those fed a lettuce diet. Snails fed a fish flake diet also grew less than lettuce-fed snails in control treatments. These results contrast with findings in the literature, where fish flake-fed snails typically grow larger than lettuce-fed snails (Berteloot et al., 2015; Besser et al., 2016; Zimmer et al., 2012). In the present study, fish flake-fed snails reached smaller sizes than expected, with freshly hatched snails growing to an average of 4 mm compared to the 10 mm reported by Berteloot et al. (2015) and Zimmer et al. (2012). While Zimmer et al. (2012) used TetraPhyll fish flakes, the current study used the same TetraMin brand fish flakes as were used in Berteloot et al. (2015). In both of these previously conducted studies, the relatively high protein content of the flakes, in comparison to lettuce, is proposed as a possible contributor to the increased growth observed in fish flake-fed snails (Berteloot et al., 2015; Zimmer et al., 2012). Since the fish flake composition in the current study should be directly comparable to that used by Berteloot et al. (2015), comparable growth would be expected. This difference may be partly attributable to organism-sourcing from different laboratory cultures, as snails used to establish the culture for the present study were sourced from VU (Amsterdam, Netherlands), while both previous studies sourced snails from INRA (Rennes, France).

Methodological factors contributing to this lower-than-expected growth may include larger test vessel volumes or a tendency for fish flakes to accumulate on the water surface, despite efforts to dissolve them. Although the portions of fish flakes in this study were equal to or greater than the amounts provided in comparable studies (Berteloot et al., 2015; Zimmer et al., 2012), the larger test vessel volumes may have diluted the fish flakes in the medium, reducing ingestion capacity for the snails. Additionally, some of the fish flakes may have adhered to the water surface, further decreasing availability to the snails.

4.3. Individual-level effects over time

Mattsson and Crémazy (2023) and Niyogi et al. (2014) concluded that Ni sensitivity in juvenile *L. stagnalis* can be reliably assessed after a three-week exposure. As noted by Crémazy et al. (2018), most chronic studies of metal toxicity in juvenile *L. stagnalis* use exposure durations shorter than 21 days. However, for some metals, full life-cycle toxicity in *L. stagnalis* has been reliably predicted using a 14-day test (Munley et al. 2013; Brix et al. 2012). The results of our single life stage tests are consistent with the need for a 21-day exposure to describe Ni toxicity to *L. stagnalis*. Although the NOEC/LOEC for freshly hatched snails did not change between day 14 and day 28 of exposure, the NOEC/LOEC decreased each week for the first 21 days of exposure in two-week-old snails (SI Figs. 5–7).

4.4. Population sensitivity over time

The first observed significant difference between Ni-exposed and control populations was a decrease in reproduction at the highest tested concentration (64.9 μg Ni L^{-1}), beginning from day 41 of population exposure (Fig. 4). By comparison, population density effects emerged more slowly, becoming statistically significant on day 119 in the highest tested concentration (Fig. 3). Once these decreases in reproduction and density emerged in the highest Ni treatment, they were observed repeatedly throughout the remainder of the exposure period. Decreased population density was also observed once in the second-highest tested concentration (7.2 μg Ni L^{-1}), at the conclusion of testing, but it is unclear whether this decrease would have been sustained with further exposure. In keeping with criteria for repeated population observations over an extended exposure period, this single observation was not considered reliable (Hommen et al., 2015). Further supporting this conclusion, no corresponding effects on reproduction were observed at this lower concentration. For the highest Ni treatment, differences in population density were almost certainly present among younger, smaller snails before day 119, but went undetected due to methodological limitations, as only snails with an estimated shell length $\geq$ 5 mm were included in population density estimates (Section 2.2.4).

An investigation of community-level Ni exposures found no significant differences in *L. stagnalis* abundance between Ni-treated (~96 μg Ni L^{-1}) microcosms and controls across two consecutive sampling events until day 56 of exposure (Hommen et al., 2015). This reported emergence of Ni effects on snail abundance occurred between the decreases in reproduction and population density observed in the population test. Because the community-level study was able to quantify the abundance of small snails, these results are consistent with the interpretation that undetected differences in the density of small snails emerged shortly after the observed decrease in population clutch production. The 11-week delay between decreased clutch production and the detected decrease in population density also aligns closely with existing knowledge regarding developmental recruitment in *L. stagnalis*. Snails enter the "growth phase" of late postlarval development about 10 weeks after hatching, at which point they typically have a shell length of approximately 3 mm (Zotin, 2009). Mollusks are known to experience high mortality during early postlarval development; consequently, population density estimates in this study may largely reflect differences in recruitment into late postlarval development stages (when Ni exposure begins during embryonic development), rather than strictly reflecting overall population density.

4.5. Individual versus laboratory population sensitivity

After 28 days of exposure (the duration of all single life stage tests), no population-level endpoints exhibited significant Ni effects. At the conclusion of testing, population-level endpoints showed sensitivity (NOECs) most comparable to that of two-week-old snails (Table 3). Population-level endpoints were more sensitive than adult individual-level endpoints, but less sensitive than freshly hatched juvenile individual-level endpoints. Although effect concentrations derived from juvenile snails, particularly freshly hatched snails, may be overly protective for population endpoints, the OECD adult reproduction test may not be protective for population-level endpoints over long-term exposures. Because the goal of ERA is generally to protect higher levels of biological organization, such as populations and communities, the relationship between individual-level sensitivity and population sensitivity warrants further scrutiny when individual-level endpoints are used to inform regulatory criteria.

Average DOC concentrations were higher in population-level testing (3.5 mg L^{-1}) than in juvenile tests (1.4 to 1.7 mg L^{-1}) (Table 1). As explored in the discussion of differences across life stages (Section 4.1), it is possible that DOC is a contributing factor to sensitivity differences between individual-level and population-level endpoints; however, the observed difference in sensitivity is too large to be explained by DOC alone. No trends in DOC concentration were observed over the course of the population experiment, with DOC remaining fairly constant (SI Figure 9). Accordingly, changing DOC over time should not be considered an explanatory factor in driving population-level dynamics under Ni exposure.

Population-level interactions, such as competition for food, evolve over time, potentially leading to the delayed manifestation of exposure effects not observed in individual-level tests. For example, food limitation becomes more likely as a population grows, which can exacerbate the effects of toxicant stress (Zimmer et al., 2012). Laboratory population endpoints may better reflect realistic joint effects of toxicant exposure and density-dependent processes, rather than direct toxicity alone. Although toxicity testing in laboratory populations is intended to incorporate ecologically relevant dynamics, such as recruitment dynamics and competition for food, key features of natural populations are absent in closed laboratory systems. In natural environments, factors such as immigration and emigration may allow behavioral compensation under food-limited conditions. Furthermore, naturally occurring interspecies interactions, such as parasitism or predator-prey dynamics, also complicate the extrapolation of single-species laboratory observations to natural environments. Genetic diversity is typically narrower in laboratory cultures, further limiting external validity. However, laboratory population exposures and the monitoring of population-level endpoints provide a tractable bridge between standard individual-level tests and the environmental scales of interest in ERA.

5. Conclusion

This study underscores the importance of considering variability across life stages in ecotoxicological assessments. Our findings suggest that standard adult-based tests for *L. stagnalis* (OECD Test No 243) may not be protective for younger life stages and population-level endpoints, highlighting the relevance of life stage-specific sensitivity data for comprehensive environmental protection. Together, these findings demonstrate the need for additional, coherent standardized juvenile test guidelines for the meaningful incorporation of *L. stagnalis* into risk assessment for Ni and other toxicants. Future research should focus on a mechanistic understanding of the differing sensitivity levels observed across life stages and their long-term implications for population- and community-level responses to Ni exposure.

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CRediT authorship contribution statement

Kristi Weighman: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Karel Viaene:** Writing –

review & editing, Supervision. **Karel De Schamphelaere:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Funding for nickel toxicity modeling was provided by NiPERA and used in support of Kristi Weighman's doctoral research.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.aquatox.2025.107612.

Data availability

Data will be made available on request.

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